

# Star Life Cycle

**Use the cue, then fade the cue.** Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

|                  |                                                                                                                                          |
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| <b>Cue focus</b> | Stars form from nebulae, spend most of their lives in the main-sequence stage, and follow different end paths depending largely on mass. |
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## Round 1 - Retrieve It

1. What is a nebula?

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2. What stage follows a protostar once stable fusion is established?

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3. What is a common end product for a Sun-like star?

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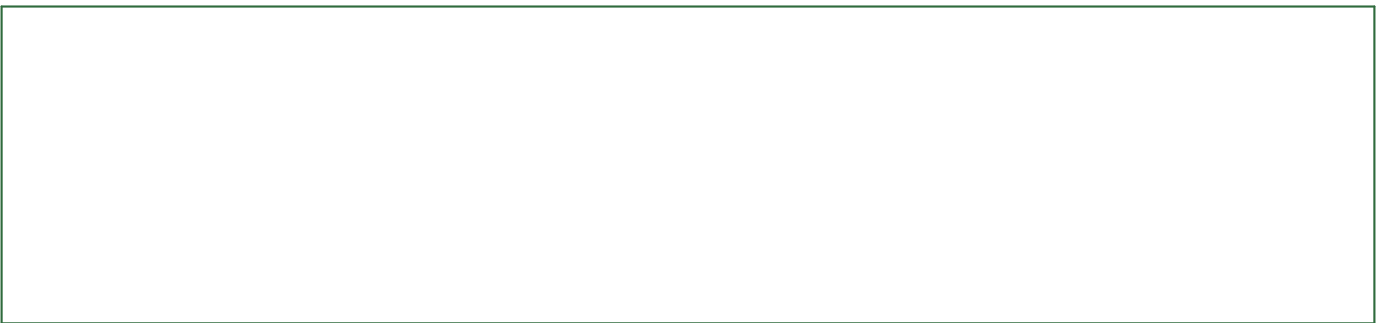
4. What major event can end the life of a massive star?

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## Round 2 - Sketch It

Draw two branching pathways from nebula to main sequence: one for a Sun-like star and one for a massive star.



# Star Life Cycle - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

## Round 3 - Apply the Relationship

1. Which factor most strongly determines which life-cycle path a star follows?

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2. What comes before a white dwarf in a Sun-like star path?

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3. What can remain after a supernova?

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## Round 4 - Reason from Evidence

Explain why the star life cycle is better shown as a branching diagram than as one single line.

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# Star Life Cycle - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

**A****Articulate**

Explain Star Life Cycle in your own words. Use one example that is not printed on the cue card.

**C****Connect**

Connect star mass to gravity. Why might a more massive star follow a more extreme end path?

**E****Extend**

Predict the broad end path of the Sun based on its being a lower-mass/main-sequence star rather than a very massive star.

**Confidence check** Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

# TEACHER KEY - Star Life Cycle

**Remove this page before student copying.** Answers below give expected ideas, not required wording.

## Retrieve It

1. A cloud of gas and dust where stars can form.
2. Main-sequence star.
3. White dwarf, after red giant and planetary-nebula stages.
4. A supernova, which may leave a neutron star or black hole.

## Sketch It - look for

Both begin nebula -> protostar -> main sequence. Sun-like: red giant -> planetary nebula -> white dwarf. Massive: red supergiant -> supernova -> neutron star or black hole.

## Apply It

1. Mass.
2. Planetary nebula after the red giant phase.
3. A neutron star or black hole, depending on mass/remnant conditions.

## Reason from Evidence - look for

Stars do not all follow the same ending; mass leads to different stages and remnants, so the sequence branches.

## ACE - Extend look-for

Red giant -> planetary nebula -> white dwarf; not a core-collapse supernova/black hole path.

### Suggested use

Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.